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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

EC4332-1 – MIXED SIGNAL VLSI DESIGN

(For EC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. The pedestal error in a Sample-and-Hold circuit is primarily caused by
 - A) Capacitor leakage during hold phase
 - B) Charge injection from the MOSFET switch when it turns OFF
 - C) Thermal noise at the input
 - D) Slew rate limitation of the op-amp
2. The magnitude of aperture error in a Sample-and-Hold circuit depends on
 - A) Slew rate of the input signal and aperture time
 - B) Charge injection from the MOSFET switch
 - C) Hold capacitor leakage
 - D) Op-amp bandwidth
3. The resolution of a DAC is determined by
 - A) Its clock frequency
 - B) The number of bits of the input digital word
 - C) The type of op-amp used
 - D) The power supply voltage
4. In an n-bit DAC, the Most Significant Bit (MSB) contributes to
 - A) The smallest voltage change in output
 - B) Half of the LSB value
 - C) The largest voltage change in output
 - D) Only digital noise
5. DAC type provides the fastest conversion?
 - A) Flash DAC
 - B) R-2R DAC
 - C) Weighted resistor DAC
 - D) Integrating DAC
6. Which of the following methods can help reduce droop error in a Sample-and-Hold circuit?
 - A) Using a larger hold capacitor
 - B) Using a switch with higher charge injection
 - C) Increasing input signal slew rate
 - D) Increasing aperture time
7. The R-2R ladder DAC requires how many resistor values for implementation?
 - A) n different resistor values
 - B) 2 resistor values
 - C) 3 resistor values
 - D) Depends on the number of bits
8. Pipelined DACs are typically used because they provide a compromise between
 - A) Low cost and low resolution
 - B) Low power and low noise
 - C) High speed and medium-to-high resolution
 - D) Flash and SAR DAC linearity
9. The main limitation of a flash ADC for high-resolution applications is____
 - A) Low speed
 - B) High latency
 - C) Exponential growth in the number of comparators
 - D) Poor linearity
10. In a pipelined ADC, each stage
 - A) Resolves all bits at once
 - B) Uses oversampling and noise shaping
 - C) Resolves a few bits and passes the residue to the next stage
 - D) Integrates the input voltage over time
11. In a dual-slope ADC, the input voltage is converted by
 - A) Direct comparison with a reference using comparators
 - B) Integrating the input over a fixed period and then de-integrating with a reference
 - C) Successive approximation of input voltage
 - D) Using oversampling and noise shaping

12. A 4-bit ADC has a reference voltage of 10 V. What is the digital output for an input of 6.2 V?
 A) 1010 B) 0110
 C) 1001 D) 0111
13. Integral Non-Linearity (INL) in an ADC refers to:
 A) Deviation of each output step from the ideal value by more than ± 0.5 LSB B) Maximum deviation of actual transfer function from ideal straight line
 C) Step size variation between adjacent codes D) The difference between actual and ideal zero input voltage output
14. Effective Number of Bits (ENOB) of an ADC is related to
 A) Resolution only B) Sampling frequency only
 C) Signal-to-Noise and Distortion Ratio D) INL error
15. In a successive approximation ADC, conversion time:
 A) Depends on the number of bits and is constant for each sample B) Is fastest among all ADC types
 C) Varies with input voltage D) Is infinite
16. In sub-micron CMOS technology, channel lengths are typically less than
 A) 1 μm B) 100 μm
 C) 10 μm D) 0.1 μm
17. The main drawback of using a MOS capacitor in sub-micron CMOS circuits is:
 A) Very high capacitance density B) Strong non-linearity with voltage
 C) High-quality factor (Q) D) Zero leakage current
18. Short-channel effects in deep sub-micron MOSFETs are mainly caused by
 A) Increased oxide thickness B) Reduction in channel length comparable to depletion width
 C) Higher threshold voltage D) Gate oxide charge
19. The main purpose of using a guard ring around sensitive analog circuits is
 A) To reduce area consumption B) To provide ESD protection
 C) To isolate circuit from substrate noise D) To reduce propagation delay
20. Shielding of interconnect lines is generally implemented using
 A) Lightly doped drain (LDD) structures B) Grounded or power supply guard lines
 C) Thicker gate oxides D) Higher threshold voltage transistors

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the working of a sample-and-hold circuit with the aid of a block diagram. With suitable waveforms, describe acquisition time and droop error in a sample-and-hold circuit	8	L*2	1	1
	b) Draw the block diagram of a pipeline DAC and briefly explain the operation. Find the output voltage for a 3-bit pipeline DAC for three cases: $D_A = 001$, $D_B = 110$, and $D_C = 101$, assume $V_{REF} = 5$ V. Also show the clock signals used for the conversion.	8	L3	2	1
2.	a) with the help of suitable waveforms, explain acquisition time, settling error and aperture error in a sample-and-hold circuit.	8	L2	1	1
	b) Draw the block diagram of a N-bit charge-scaling DAC and briefly explain the same. Determine the analog output voltage in a 3-bit charge scaling DAC with $V_{REF} = 5$ V, $C = 0.5$ pF, for digital input, $D_2D_1D_0 = 010$ and 101 .	8	L3	2	1
3.	a) With the help of relevant waveforms, explain droop error and aperture time in a sample-and-hold circuit. Also determine the maximum sampling error in a S/H circuit for a sinusoidal input $V_{IN} = A \sin 2\pi f t$ where A is 2 V and $f = 100$ kHz. Assume the aperture uncertainty is 0.5 ns.	8	L2	1	1

- b) With reference to DAC, explain Integral Non-Linearity and Most Significant Bit. In a R-2R DAC, $2R = 2\text{ k}\Omega$ and $V_{REF} = 5\text{ V}$. Determine the analog output for the digital input, $D_2D_1D_0 = 100$. Draw the structure of the same.

8 L3 2 1

Unit – II

4. a) With reference to an ADC, describe the various errors that occur due to variations in slope, unequal step sizes, and displacement of the transfer characteristic from its ideal position.
- b) Illustrate the conversion process in a 3-bit pipelined ADC for $V_{in} = 2\text{ V}$, considering $V_{REF} = 5\text{ V}$. Also, briefly describe the working principle of a pipelined ADC.

8 L2 3 1

8 L3 4 1

5. a) With reference to ADC explain:

- Gain error
- Quantization error
- Integral non-linearity

8 L2 3 1

- b) Draw the block diagram of a SAR ADC and briefly explain its conversion steps. Also, describe the binary search algorithm used in the conversion process in a 3-bit SAR ADC for $V_{in} = 2.5\text{ V}$ considering $V_{REF} = 8\text{ V}$.

8 L3 4 1

6. a) Draw and clearly explain individual ADC transfer characteristics highlighting the following types of errors:

- Missing codes
- Differential non-linearity
- Offset error

8 L2 3 1

- b) With the help of a block diagram of N-bit flash ADC, explain the conversion principle. Also, design a 2-bit Flash ADC in which the maximum power dissipation in the resistor ladder is 2.5 mW , with $V_{REF} = 5\text{ V}$.

8 L3 4 1

Unit – III

7. a) Explain the operation of a TSPC-based delay element with diagrams.
- b) List the desired characteristics of Capacitors and also explain Floating MOS Capacitors.

8 L2 5 1

8 L2 5 1

8. a) Discuss the important VLSI mixed-signal layout strategies for floor planning and designing fully differential circuits to minimize noise and ensure signal integrity.

8 L2 5 1

- b) Describe the functioning of a delay element implemented using pass transistors and CMOS inverters, with a diagram.

8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
